

Chicken McNugget Theorem

The **Chicken McNugget Theorem** (or **Postage Stamp Problem** or **Frobenius Coin Problem**) states that for any two **relatively prime positive integers** m, n , the greatest integer that cannot be written in the form $am + bn$ for **nonnegative** integers a, b is $mn - m - n$.

A consequence of the theorem is that there are exactly $\frac{(m-1)(n-1)}{2}$ positive integers which cannot be expressed in the form $am + bn$. The proof is based on the fact that in each pair of the form $(k, mn - m - n - k)$, exactly one element is expressible.